

Core Backend Interview Questions (REST, JSON, Middleware, Auth)

REST APIs (GET, POST, PUT, DELETE)

1. What is a REST API? What are its key principles?
2. Difference between **GET, POST, PUT, PATCH, DELETE**
3. Is GET request idempotent? What does idempotent mean?
4. When should you use PUT vs PATCH?
5. Can POST be idempotent? Explain with example
6. What are HTTP status codes? Give examples (200, 201, 400, 401, 404, 500)
7. What is the difference between 401 and 403?
8. How do you design a RESTful URL for:
 - Get all users
 - Get user by ID
 - Delete a user

Request-Response Cycle

9. Explain the complete request-response lifecycle
10. What happens when you type a URL in the browser?
11. What are headers in HTTP request and response?
12. What is the difference between query parameters and path parameters?
13. What is statelessness in HTTP? Why is it important?

JSON Handling

14. What is JSON? Why is it widely used?
15. Difference between JSON and XML

Middleware

19. What is middleware in backend frameworks?
20. Give examples of middleware (logging, authentication, error handling)
21. What is the order of middleware execution?
22. Difference between middleware and controller
23. How would you implement a logging middleware?

JWT (JSON Web Token)

24. What is JWT and how does it work?
25. What are the parts of JWT (Header, Payload, Signature)?
26. How is JWT different from session-based authentication?
27. Where is JWT stored on the client side?
28. What are the disadvantages of JWT?

Session-Based Authentication

29. What is session-based authentication?
30. How are sessions stored and managed?
31. Difference between cookies and sessions
32. What is session hijacking?
33. How do you secure session-based authentication?

Client → Server → Database Flow

1. Explain the complete flow: Client → Server → Database → Response
2. What happens when a user clicks a button in a web app?
3. How does the backend communicate with the database?
4. What is the role of an API in this flow?
5. Where does validation happen: client, server, or both?
6. What are common bottlenecks in this flow?
7. How do you handle failures between server and database?

MVC Architecture

8. What is MVC architecture? Explain each component
9. What is the role of Model, View, and Controller?
10. Why is MVC used in backend systems?
11. Difference between MVC and layered architecture
12. Can you explain MVC using a real-world example?
13. What are the disadvantages of MVC?

Monolith vs Microservices

14. What is a monolithic architecture?
15. What are microservices?
16. Monolith vs Microservices – key differences
17. When should you choose monolith over microservices?
18. What are challenges in microservices architecture?
19. What is service-to-service communication? (REST, messaging)
20. What is API Gateway in microservices?

API Design Principles

21. What makes a good API design?
22. What are RESTful best practices?
23. How do you version APIs? (v1, v2, headers)
24. How do you handle errors in API design?
25. What is pagination and why is it needed?
26. What is rate limiting? Why is it important?

Scalability Basics (VERY IMPORTANT)

27. What is scalability? Difference between vertical and horizontal scaling
28. What is load balancing?
29. What is caching? Where can caching be applied?
30. What is database indexing?
31. What are read-heavy vs write-heavy systems?
32. How would you scale a backend serving millions of users?
33. What is CDN and why is it used?

OOPS Concepts-

What are the main features of OOPs?
 What are access specifiers and what is their significance?
 What are the various types of inheritance? Explain.
 What is meant by polymorphism? Write its types.
 What is the difference between overloading and overriding?
 What is data abstraction?
 What is an abstract class?
 What is a class? What is an object?
 What is encapsulation?
 What is a constructor?
 What are the various types of constructors?
 What is a copy constructor?
 What is a destructor?
 What is the difference between a structure and a class in C++?
 What are friend functions?
 What is the virtual function and pure virtual function?
 C++ Static Data Members
 What is enumeration?
 What is this pointer?

Programming language proficiency in python/Java

Python

- What are the key differences between list, tuple, set, and dict?
1. What is mutability? Which Python data types are mutable and immutable?
 2. Explain shallow copy vs deep copy with examples
 3. What is the difference between is and ==?
 4. How is memory managed in Python? (garbage collection, reference counting)
- What are *args and **kwargs?
6. What are lambda functions? When should you use them?
 7. What are decorators? Give a practical use case
 8. What is recursion? Write a recursive function example
 9. What is the difference between map(), filter(), and reduce()?
- Why are sets faster than lists for lookup operations?
11. Can dictionary keys be mutable? Why or why not?
 12. What is the time complexity of:
 - list append
 - dict lookup
 - set add
 14. Difference between defaultdict and normal dictionary
 15. How does hashing work in Python dictionaries?

Java

1. What is the difference between **JDK, JRE, and JVM**?
2. What are the main features of Java (OOP concepts)?
3. Difference between **method overloading** and **method overriding**
4. What is **encapsulation** and why is it important?
5. What is **abstraction**? Difference between abstract class and interface
6. How does **garbage collection** work in Java?
7. What is the difference between **stack memory and heap memory**?
8. What is the **String pool** in Java?
9. Difference between String, StringBuilder, and StringBuffer
10. Why are strings immutable in Java?
11. Difference between ArrayList and LinkedList
12. Difference between HashMap and HashSet
13. How does HashMap work internally?
14. What is **hashing** and how are collisions handled?
15. What is the **load factor** and rehashing in HashMap?
16. Difference between HashMap and TreeMap
17. Difference between HashSet and TreeSet
18. What is Comparable vs Comparator?
19. How do you sort objects in Java?
20. What happens if two keys have the same hashCode?

DBMS

- What is the difference between WHERE and HAVING?
- Write a query to fetch the second highest salary from a table.
(Using subquery / Limit offset / Rank)
- Inner Join / Left Join / Right Join / Outer Join
- Find the number of employees in each department.
- What are keys (Primary key, Candidate Key, Alternate Key, Foreign Key, Composite key, Super key)
- What are PRIMARY KEY and UNIQUE constraints?
- What is a VIEW in SQL?
- Write a query to fetch top 3 highest-paid employees.
(Use limit offset / Rank)
- Difference between DELETE, TRUNCATE, and DROP?
- What is an entity-relationship diagram (ERD)?
- What is normalization? Why is it important? What are the various forms of Normalization?
- Explain different types of relationships amongst tables in a DBMS.
- What are tuples and attributes in DBMS?
- Find employees who earn more than the average salary.
- Difference between UNION and UNION ALL?
- What is the DISTINCT keyword in SQL?
- What is the purpose of the GROUP BY clause in SQL?

- What are indexes? What is the difference between a clustered and non-clustered index?
- What are ACID properties in a transaction?
- What is Pattern Matching in SQL? (Wildcard)
- What are the differences between OLTP and OLAP?
- **DDL (Data Definition Language):** CREATE, ALTER, DROP, TRUNCATE
- **DML (Data Manipulation Language):** SELECT, INSERT, UPDATE, DELETE
- **DCL (Data Control Language):** GRANT, REVOKE
- **TCL (Transaction Control Language):** COMMIT, ROLLBACK, SAVEPOINT

NoSQL (MongoDB)

- What is NoSQL? How is it different from SQL databases?
- What is MongoDB? Why is it widely used?
- What is a document and collection in MongoDB?
- What is BSON? How is it different from JSON?
- What is schema-less design? What are its advantages?
- How do you perform basic operations in MongoDB? (insert, find, update, delete)
- What is _id in MongoDB?
- What are indexes in MongoDB? Why are they important?
- What is the difference between embedded and referenced documents?
- What is sharding and replication in MongoDB?

SaaS Basics

1. What is SaaS (Software as a Service)?
2. What are the advantages of SaaS?

Multi-Tenant Architecture

6. What is multi-tenant architecture?
7. Difference between single-tenant and multi-tenant systems

Deployment Basics

11. What is deployment in backend systems?
12. What are the steps to deploy a backend application?
13. What is CI/CD?

Docker

15. What is Docker? Why is it used?
16. What is a Docker container? How is it different from a virtual machine?
17. What is a Docker image?
18. What is a Dockerfile?

Hosting & AWS Basics

19. What is cloud computing?
20. What are basic AWS services used in deployment? (EC2, S3, RDS)

Git Basics

1. What is Git? Why is it used?
2. What is the difference between Git and GitHub?
3. What is a repository in Git?

4. What is a commit?
5. What is the difference between git add and git commit?
6. What does git push do?
7. What does git pull do?
8. Difference between git fetch and git pull
9. What is a branch in Git?
10. What is merging in Git?

GitHub Usage

11. What is GitHub?
12. What is a pull request (PR)?
13. What is a fork in GitHub?
14. What is the difference between fork and clone?
15. How do you resolve merge conflicts?
16. What is GitHub Actions (basic idea)?
17. You made changes but haven't committed—how do you check status?
18. You committed code but want to undo it—what will you do?

Data Structures-

What is linear and non linear data structure?

What is a stack data structure? What are the operations available on stack?

What is a Queue data structure? What are the operations available on stack?

What is array data structure?

What is a linked list and its types?

What are the advantages of a linked list over an array? In which scenarios do we use Linked List and when Array?

What is an asymptotic analysis of an algorithm?

What is a hashmap in data structure? and its time complexity.

What is binary tree data structure?

What is binary search tree data structure?

What are tree traversals? Inorder, preorder, postorder.

What is a priority queue?

What is graph data structure and its representations?

What is the difference between the Breadth First Search (BFS) and Depth First Search (DFS)?

What is AVL tree data structure

Define Red-Black Tree

What is a heap data structure?

Which sorting algorithm is considered the fastest? Why?

Learn best case, average case, worst case time complexity of Bubble sort, selection sort,

insertion sort, merge sort, quick sort.

What is the merge sort? How does it work?

What are the advantages of a binary search over a linear search?

What is the time complexity of binary search?

What is hashing? What are the different techniques of hashing.

What is the time complexity of search, insert, and delete operations in a Hash Table?

What is recursion?

What is the time complexity to search in an unsorted and sorted array?

What is time complexity of Linked List operations? (insertion, deletion, search, traversal)

What approach would you take to find the shortest route between two locations? (Dijkstra algorithm)

What is the minimum spanning tree?